

Chris Zhou

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EDUCATION

University of Southern California <i>Master in Computer Science</i>	2023 – 2025
George Mason University <i>Bachelor in Computer Science (Dean's List)</i>	2017 – 2021

WORK EXPERIENCE

Jianzhao Enterprise Management Consulting Services Co., Ltd.	Jun 2024 – Sep 2024
Software Engineer Intern	Remote

- Developed business consulting and management support platform using **Spring Boot** and **PostgreSQL**, supporting 1.1k+ users with complex multi-criteria business data queries and paginated results.
- Optimized database performance by consolidating multiple SQL queries into a single indexed query, reducing average response time from **~3s to ~800ms**.
- Designed real-time dashboards by replacing HTTP long polling with WebSocket-based communication, reducing message loss by **~70%** under concurrent load.
- Deployed services on Alibaba Cloud with auto-scaling and load balancing, improving system availability and reducing peak-load latency.

Right Bank Development Co., Ltd.	May 2023 – Aug 2023
Software Engineer Intern	Remote

- Contributed to a microservices-based order management system using Spring Boot, Eureka, and Spring Cloud Gateway, supporting 6,500+ daily orders.
- Implemented **MySQL sharding** and date-based partitioning with ShardingSphere, reducing single-table size from **~100K** to **~10K** rows and improving query performance by **~65%**.
- Built a high-throughput order processing pipeline using **Kafka** to ensure reliable message delivery, reducing message loss by over 90% during traffic spikes.
- Implemented authentication and RBAC using Spring Security and **OAuth 2.0** to secure multi-tenant access.

PROJECTS

Environmental Data Monitoring Platform	Jul 2025 – Present
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- Designed and implemented an environmental monitoring system on **AWS** (EC2, S3, RDS), ingesting data from 200+ monitoring points with real-time dashboards and alerting.
- Improved system resilience using **Redis** caching, **Redisson** distributed locks, and async processing, supporting 1,000+ concurrent requests.
- Built an asynchronous data ingestion pipeline with **RabbitMQ** and dead-letter queues, ensuring reliable delivery and retry handling.
- Optimized frontend performance with route-based lazy loading, reducing initial bundle size from 1.8MB to 850KB.

SKILLS

- Languages:** Java, Python
- Frameworks:** Spring Boot, Spring Cloud
- Databases & Cloud:** MySQL, PostgreSQL, Redis, AWS (EC2, S3)
- Tools & Others:** Docker, Git, Nginx, RESTful APIs
- Messaging & Distributed Systems:** RabbitMQ, Redisson, Kafka